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REVIEW

Cold-Induced Hypertension as Life-Environment Disease in Winter: Focus on Data From Japan

Kazuomi Kario

ABSTRACT: An increase in the rate of cardiovascular events (eg, myocardial infarction, stroke, heart failure) during the winter season has been reported worldwide, including Japan. As 1 contributor to the increase in cardiovascular risk during the colder months, winter hypertension likely reflects an interaction between environmental factors and human physiological responses. In particular, the prognostically important morning blood pressure (BP) surge is accentuated in winter versus other seasons, as is BP variability. Some individuals may exhibit more marked changes in BP in response to cold exposure, referred to as thermosensitive hypertension. During winter mornings, sympathetic activation due to cold stress and the arousal response overlap, producing a synergistic effect that raises baseline BP, amplifies BP variability, and augments the morning BP surge simultaneously. These mechanisms help explain why cardiovascular event risk peaks during the early morning hours in winter. Approaches to optimizing the living environment and lifestyle during winter to help reduce cold-induced increases in BP are discussed, with a focus on Japan. Incorporation of home BP monitoring and newer approaches, such as digital therapeutics are also important. Overall, winter BP management should consider BP variability, time of day (chronobiology), and the environment, rather than focusing solely on absolute BP levels. The period immediately after awakening represents the most dangerous time window, during which cold exposure, low indoor temperature, awakening-related sympathetic activation, and initiation of physical activity converge. Effective management for individual cardiovascular risk reduction requires a comprehensive approach that optimizes living conditions, lifestyle factors, BP monitoring, and hypertension pharmacotherapy.

Key Words: blood pressure ■ cardiovascular diseases ■ cholesterol ■ heart failure ■ myocardial infarction

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WINTER-ASSOCIATED INCREASE IN CARDIOVASCULAR RISK

Although the magnitude of seasonal influence varies between different regions and countries around the globe, numerous epidemiological studies have shown that the occurrence of cardiovascular events, such as myocardial infarction, stroke, and heart failure, significantly increases during the winter season.[1–3] The seasonal rise in mortality from cardiovascular events has been associated with lower indoor and outdoor temperatures.[4] In Japan, the rate of major cardiovascular diseases, including myocardial infarction, stroke, aortic dissection, and heart failure, is ≈1.5-fold higher in December and January than in the summer months (Figure 1).[5]

Overall, winter constitutes a high-risk season in which the interaction between meteorological conditions and physiological responses amplifies an individual's risk of experiencing a cardiovascular event. Although seasonal increases in factors such as cholesterol level, blood sugar, body mass index, and waist circumference can also contribute to the winter-associated increase in cardiovascular risk,[6] this article focuses on changes in blood pressure (BP) associated with cold temperatures in winter as a contributor to individual cardiovascular risk in Japan.

THERMOSENSITIVE HYPERTENSION

Studies have reported that BP fluctuates in response to meteorological factors, such as temperature, humidity,

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Nonstandard Abbreviations and Acronyms

ABPM	ambulatory BP monitoring
BP	blood pressure
DTx	digital therapeutics
HBPM	home BP monitoring
MBPS	morning BP surge
RVLM	rostral ventrolateral medulla
SBP	systolic BP
TRPM8	transient receptor potential melastatin 8

and wind speed, and that cold-induced BP elevation is clinically significant and cannot be ignored.[7–9] The early morning hours during winter represent the highest-risk time for the cardiovascular system.[10] BP has a clear circadian rhythm, characterized by a rapid rise after awakening (the morning BP surge [MBPS]), and all major cardiovascular events occur more frequently in the morning, especially in the elderly.[11,12] Morning home BP is well known to be strongly associated with cardiovascular risk.[13–17]

During winter, all BP parameters increase, including office BP, home BP, and ambulatory BP.[18–27] With respect to home BP, the winter-related increases are greater for morning BP than for bedtime BP.[18–20,22,24] Several patterns of seasonal BP change have been identified in individuals being treated with antihypertensive drug therapy: those with markedly higher home BP in winter compared with summer, and those with paradoxical decreases in winter BP (abnormal seasonal variation), had higher cardiovascular event rates than people with smaller seasonal BP variations.[28]

Kario et al[19] defined the term thermosensitive hypertension to describe a patient phenotype characterized by an exaggerated BP response to cold exposure (Figure 2). This is particularly relevant for colder indoor temperatures, as indicated by data showing that colder regions in Japan that have better insulated housing had lower rates of cardiovascular deaths in winter than warmer areas that have less well-insulated housing. Data from the prospective, cross-sectional Japanese Nationwide Smart Wellness Housing survey of 3775 participants (2095 households) showed that a 10°C decrease in indoor temperature raised morning home systolic BP (SBP) by ≈8.2 mm Hg.[30] SBP in older individuals and women was reported to be particularly sensitive to changes in indoor temperature.[30]

Even when office BP is well controlled, masked morning hypertension (defined by uncontrolled morning home BP) becomes more prevalent during winter.[22] Furthermore, morning home BP is more strongly associated with winter-onset cardiovascular disease than bedtime home BP.[31] This highlights the importance of home BP monitoring (HBPM) to detect the exaggerated MBPS (especially in winter). To encourage this, the Japanese Society of Hypertension has issued the Declaration for the Eradication of Morning Hypertension,[32] and launched a nationwide public campaign (the Asakatsu

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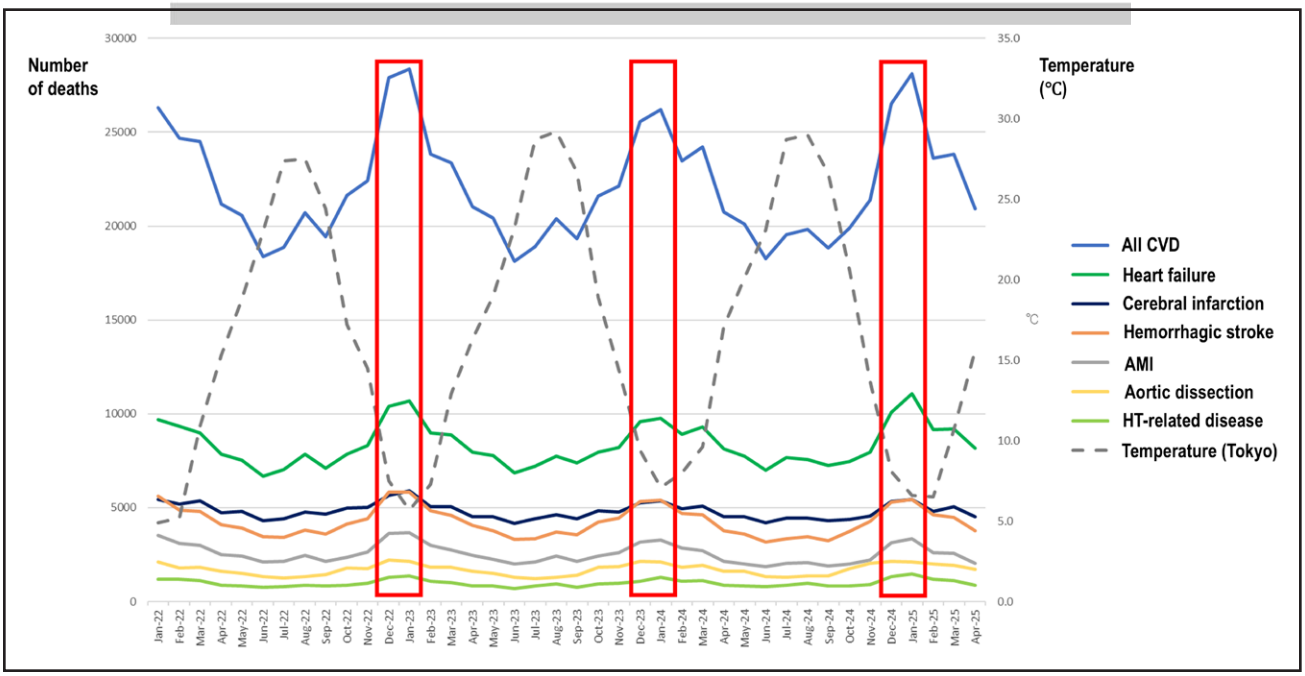

Figure 1. Seasonal variation in cardiovascular death in Japan.
Source: Ministry of Health, Labor and Welfare: Monthly Vital Statistics Report (Preliminary Data), and Japan meteorologic Agency: Monthly Mean of Daily Average Temperature (°C). Red boxes indicate peaks in cardiovascular death during the months of December and January. AMI indicates acute myocardial infarction; CVD, cardiovascular disease; and HT, hypertension. Reproduced from Kario et al.[5]

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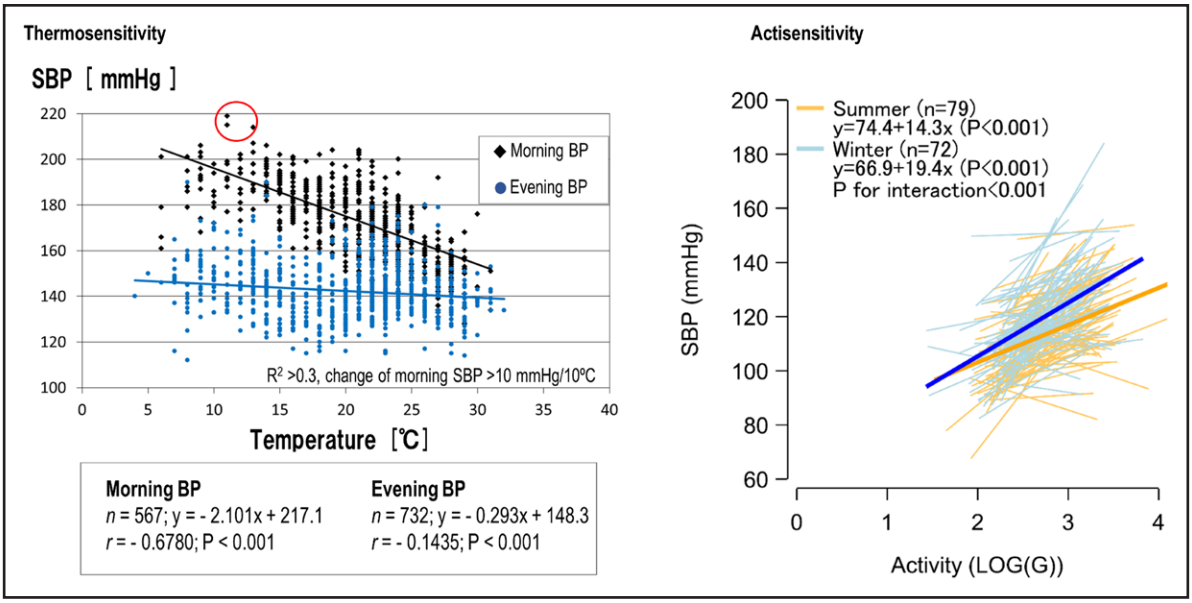

Figure 2. Thermosensitivity and seasonal variation of actisensitivity.
The graph on the left shows the relationship between environmental temperature at the time of blood pressure (BP) measurement and systolic BP (SBP) for 1 patient over 1 year; the slope of the line for the relationship between increasing temperature and SBP (thermosensitivity) is steeper in the morning than in the evening. The red circle highlights outliers who have particularly high morning SBP at low temperatures. Thermosensitive hypertension is defined as a 10-mm Hg increase in SBP per 10°C decrease in temperature. The graph on the right shows the relationship between physical activity in the 5 minutes before BP measurement (measured using actigraphy; reproduced, with permission, from Kario K. 2018[19]) and SBP in winter (blue lines) and summer (orange lines; each line represents data from an individual patient). The dark blue and dark orange lines show the average slope of data from all patients for winter and summer SBP values, respectively. Actisensitivity is greater in winter than in summer (reproduced, with permission, from Kario K. 2017[29]).

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BP initiative) to promote self-monitoring of morning BP in diverse settings (eg, home, workplace, pharmacy, gym, public facilities).[33–35]

INCREASED BP VARIABILITY IN WINTER

Winter is also characterized by an increase in BP variability. With respect to morning home BP measurements, nighttime-morning indoor temperature fluctuations (morning-evening difference) and day-to-day temperature variability were significantly associated with circadian home BP variability and day-by-day BP variability indices, including SD, coefficient of variation, average real variability, and variability independent of the mean.[9,36] These increases in home BP variability are known cardiovascular risk factors,[37–39] and greater BP variability during winter has been shown to be associated with target organ damage.[40]

In addition to resting home BP, cold exposure also amplifies dynamic BP variability assessed using ambulatory BP monitoring (ABPM). ABPM studies have documented the occurrence of a pronounced MBPS during winter.[21,23] Decreases in indoor temperature and ambient indoor/outdoor temperatures have been shown to correlate with a greater morning BP surge.[41,42] In a study examining distal (wrist/ankle) and proximal (abdominal) skin temperature together with ABPM, lower distal skin temperature was associated with higher daytime ambulatory BP.[43]

Using ABPM equipped with an accelerometer,[44] the HI-JAMP study found that the magnitude of the morning BP surge is closely associated with actisensitivity—the reactivity of BP to physical activity.[45] In a cohort of ambulatory patients at cardiovascular risk, within-individual comparison between summer and winter showed that actisensitivity significantly increased in winter.[29] This indicates that the same amount of physical activity induces disproportionately greater BP elevation during the winter season (Figure 3). The documentation of this phenomenon supports the concept that cold exposure and reductions in indoor temperature potentiate BP reactivity. Increased actisensitivity was particularly marked among older adults and was linked to a more pronounced morning BP surge.[45]

During winter, it is therefore possible that a combination of environmental stressors, including decreases in indoor and outdoor temperatures, cold exposure, and reduced physical activity, act synergistically to increase baseline BP and amplify BP variability, contributing to cold-induced MBPS.[24,47,48] In older adults with impaired BP regulatory mechanisms living in regions where houses are not well insulated or heated overnight, indoor temperature differences encountered when moving from the bedroom to cooler areas such as the bathroom or toilet could combine with awakening to stimulate sympathetic activity and vasoconstriction.[24] This combination of pressor stimuli is consistent with the resonance

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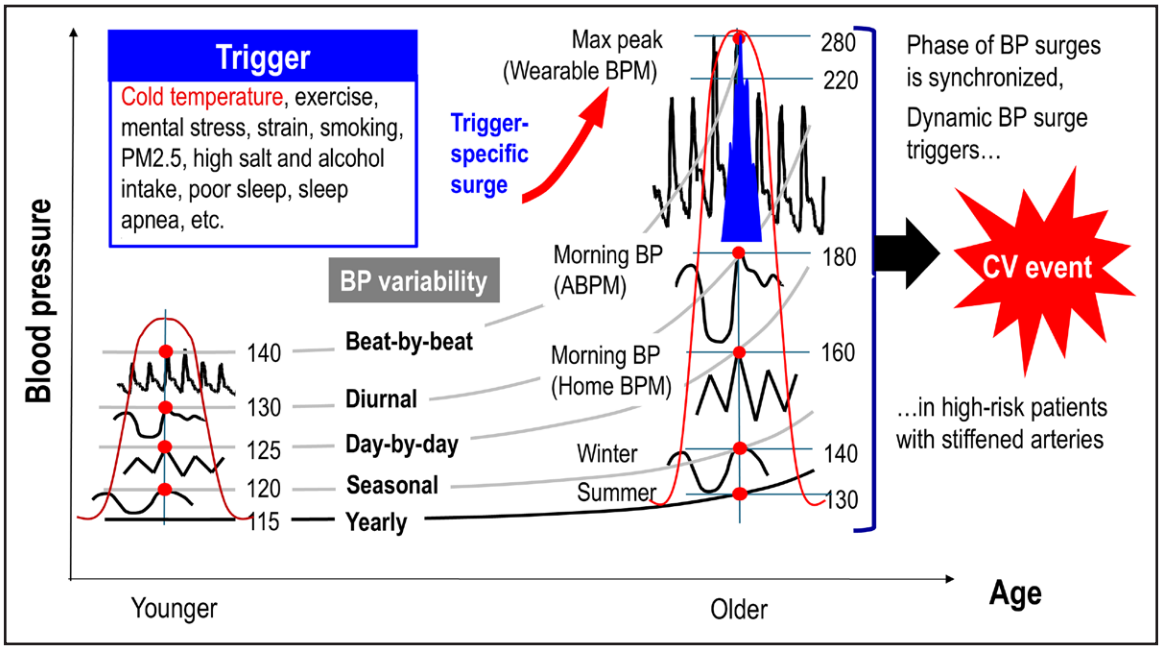

Figure 3. Resonance hypothesis of blood pressure (BP) surge to trigger cardiovascular (CV) events.
ABPM indicates ambulatory blood pressure monitoring; BPM, blood pressure monitoring; and PM2.5, fine particulate matter (airborne particles with a diameter of ≤2.5 μm). Reprinted from Kario et al[46] with permission. Copyright ©2015, Oxford University Press.

hypothesis of BP surge (Figure 3). The resonance hypothesis proposes that when the peaks of different types of BP surge/variability occur simultaneously, there is a pathologically large surge in BP that can trigger an acute cardiovascular disease event.[46,49] From the perspective of 24-hour BP control in areas where indoor overnight temperatures are much lower during winter, the morning period should be regarded as a critical high-risk window for this type of resonance.

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POTENTIAL MECHANISMS UNDERLYING COLD-INDUCED BP CHANGES IN WINTER

Exposure to cold outdoor temperatures and fluctuating indoor temperatures due to factors such as poor insulation, housing design, and lack of overnight heating can contribute to increased BP and BP variability via a variety of mechanisms. In the context of BP and hypertension, winter-associated increases in BP and BP variability are likely to result from a combination of physiological cold-defense responses and winter-specific environmental and behavioral factors.

From a physiological perspective, cold exposure can trigger activation of the central autonomic nervous system and sympathetic excitation (Figure 4).[50] Cold stimuli are first detected by TRPM8 (transient receptor potential melastatin 8) cold receptors, which are densely distributed in the extremities.[51,52] Afferent signals are transmitted through the spinal cord and medulla to the posterior hypothalamus. The hypothalamus integrates this input as cold stress and initiates simultaneous sympathetic activation to maintain core body temperature.[53] The hypothalamic output is relayed to the rostral ventrolateral medulla (RVLM), where increased firing of RVLM neurons leads to a rapid surge in global sympathetic outflow.[54,55] This results in peripheral vasoconstriction, increased systemic vascular resistance, and elevations in heart rate and myocardial contractility driven by augmented norepinephrine secretion, culminating in an abrupt BP surge and progressive cold-induced sympathetic tonus leading to sodium retention.[53] Cold exposure also activates central noradrenergic pathways via the locus coeruleus[56,57] and increases cortisol secretion via the hypothalamic-pituitary-adrenal axis,[58] further amplifying sympathetic responsiveness.[59]

In contrast, the MBPS is primarily initiated by circadian and arousal-related autonomic activation mediated by the suprachiasmatic nucleus of the hypothalamus.[15] Awakening is accompanied by increased activity of the locus coeruleus-norepinephrine system,[60] activation of the cortisol awakening response via the hypothalamic-pituitary-adrenal axis,[61] and enhanced hypothalamic input to the RVLM.[62] These processes collectively cause a rapid elevation in sympathetic output and BP.[15] Thus, the MBPS does not represent a reactive decrease in BP but rather a proactive, coordinated sympathetic activation arising from the hypothalamus-locus coeruleus hypothalamic-pituitary-adrenal axis-RVLM arousal network.[15,63]

As described above, cold stress and the arousal response in the morning stimulate the hypothalamus, locus coeruleus, hypothalamic-pituitary-adrenal axis, and RVLM in the same direction, converging to enhance sympathetic activity. Therefore, to explain the exaggerated winter morning surge in BP, we propose the mechanistic

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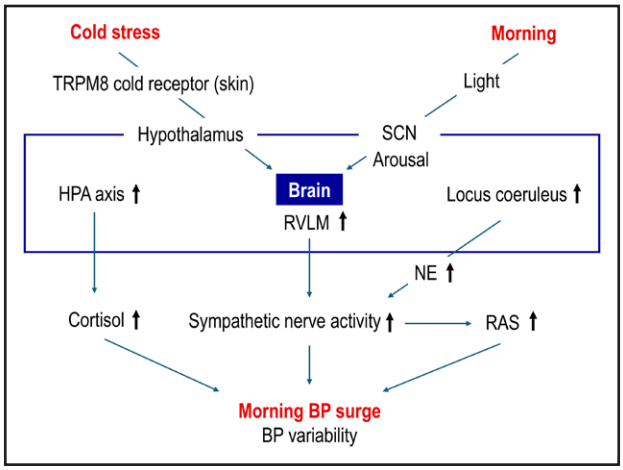

Figure 4. Mechanisms underlying the winter morning surge in blood pressure (BP).
HPA indicates hypothalamic-pituitary-adrenal axis; NE, norepinephrine; RAS, renin-angiotensin system; RVLM, rostral ventrolateral medulla; SCN, suprachiasmatic nucleus; and TRPM8, transient receptor potential melastatin 8.

cold-awakening environment-lifestyle hypothesis, which states that during winter mornings, sympathetic activation by cold stress, and sympathetic activation by the arousal response overlap, producing a synergistic effect that raises baseline BP, amplifies BP variability, and augments the morning BP surge simultaneously (as per the resonance hypothesis described above). These pathophysiological considerations, even though partly hypothetical, remind us to consider hypertension not only as a lifestyle disease but also as an environmental disease.[63] In older adults, reduced baroreflex sensitivity and impaired thermoregulation, and reduced insulation and heat production in lean individuals, make these groups particularly susceptible to cold-induced morning BP responses.[23,64]

Direct genetic evidence to support TRPM8 as a causal mechanism for thermosensitive hypertension or cold-associated morning BP surge is limited. In a genome-wide association study of BP responses in 1881 Han Chinese, a variant near TRPM8 (rs7577262) was genome-wide significantly associated with the SBP response to the cold pressor test,[65] but this reflects acute cold-stress reactivity and does not establish causality. The study also noted that most identified variants were not functional and suggested that deep sequencing is needed to identify rare/low-frequency functional variants, highlighting an evidence gap that omics approaches (eg, functional genomics, proteomics, metagenomics) may help to address.

STRATEGIES TO ATTENUATE THE IMPACT OF COLD INDOOR TEMPERATURES ON BP

Optimization of the Living Environment

When managing winter BP, optimizing the microenvironment (including housing, indoor temperature, clothing, and daily activity patterns) is critically important (Table). As a life-environment modification, retrofitting insulation and maintaining a warm indoor temperature has been shown to significantly reduce BP.[63,64] Furthermore, the BP-lowering effect of retrofitted insulation and maintaining a warm, constant indoor temperature was greater in people with hypertension, a higher salt intake, a lower body mass index, current smokers, and those who consumed alcohol.[63,64] In addition, data from a randomized controlled trial showed that intensive room heating (increasing room temperature by 10 °C) has been shown to decrease morning BP and the morning BP surge in winter compared with a cooler room.[66]

The World Health Organization recommends that indoor room temperature be maintained at ≥18 °C[67] to prevent cardiovascular heat shock. This recommended indoor temperature threshold might be better to be up to 22 °C for elderly individuals because a significant association between cooler room temperature and an increase in morning SBP has been documented up to around 22 °C.[30]

Although environmental modification is a core component of winter hypertension management, the specific environmental modifications needed will vary between regions in countries. For example, in Japan, where room walls are thin, structural adjustments may be necessary, whereas increasing the heating before waking may be sufficient in places where houses have thicker walls/better insulation.

Lifestyle Modifications

As well as increasing the temperature of the indoor environment, it may also be useful to institute other simple lifestyle changes (Table). This could include wearing warm clothing such as a hat, which has been shown to reduce the BP response to cold exposure.[68] Drinking a warm beverage could also be beneficial because having a hot drink immediately after awakening may help attenuate increases in blood viscosity and the MBPS because determinants of blood viscosity (eg, hematocrit and fibrinogen) increase in the morning.[69] Light stretching or walking might mitigate excessive sympathetic activation,[70] which could also attenuate the MBPS.[71,72] This is especially relevant in winter because this is a time when physical activity tends to decline.[73,74] Exposure to cold temperatures during the day has been associated with increased salt intake, as reflected by increases in urinary sodium excretion,[75] which contributes to nocturnal BP elevation and reduced nighttime BP dipping.[75] Therefore, strict salt reduction may be particularly effective during winter.

Digital therapeutics (DTx) that support the implementation and maintenance of lifestyle interventions (including increasing physical activity and reducing salt intake) represent an emerging option for the management of

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Table. Suggested Approaches to Help Optimize Blood Pressure Management in Winter

Category	Specific actions	Expected effects (BP/CV)
Environmental modification		
Indoor temperature control (evidence level B)	Maintain living and bedroom temperatures at ≥18 °C (ideally ≥22 °C in the elderly)	Suppress cold-induced rapid BP rise (morning surge)
	Avoid completely turning off heating at night to prevent morning temperature drops.	Prevent excessive sympathetic activation.
Thermal management of indoor pathways (evidence level B)	Warm pathways from the bedroom to the toilet/bathroom.	Reduce BP surge caused by vasoconstriction due to temperature differences.
	Begin morning activities in a warm room.	
	Use socks, indoor footwear, and warm clothing.	
Improved insulation and heating strategies (evidence level B)	Improve insulation of windows/walls, and prevent drafts	Lower morning home BP and reduce BP variability
	Use timer heating or low-level heating at night.	Greater benefit in older adults, people with a low BMI, smokers, drinkers, and individuals with hypertension
Lifestyle modification		
Warm clothing (evidence level B) and a warm drink on waking (evidence level C)	Use socks, indoor footwear, and warm clothing	Prevent abrupt peripheral vasoconstriction
	Apply mild warm compresses (eg, hot packs) to hands/feet	Reduce blood viscosity
	Have a warm drink immediately after waking.	
Morning routine (asakatsu BP; evidence level B)	Measure home BP every morning on waking	Early detection of winter BP elevation and variability
	Confirm home BP <130 mm Hg	Decrease home BP
	In-bed stretching, deep breathing, limb exercises	Suppress sympathetic overactivation
		Mitigate combined cold and awakening stress.
Medication adherence (evidence level A)	Take antihypertensive medications consistently	Better treatment adherence
	Adjust therapy in consultation with health care providers.	Better BP stability during winter
Diet (evidence level A) and hydration (evidence level C)	Limit salt in hot-pot dishes and soups	Prevent winter salt-related BP elevation
	Increase intake of potassium-rich vegetables and fruits.	Maintain endothelial function
	Hydrate after waking and before bedtime.	Prevent hidden dehydration- and viscosity-related BP increases.
Regular physical activity (evidence level A)	Indoor walking, stair climbing	Improve vascular reactivity.
	Radio calisthenics, light strength training	Reduce excessive sympathetic responses.
		Blunt morning BP surge
Sleep and circadian rhythm (evidence level B)	Warm the bedroom before sleep	Prevent nighttime BP elevation and morning hypertension
	Maintain consistent sleep-wake times.	
	Ensure exposure to sunlight or bright light during the daytime	Reduce sympathetic load via circadian rhythm stabilization
		Improve sleep quality and BP stability.
Digital therapeutics and lifestyle management (evidence level B)	Use digital tools to support behavior change and lifestyle modifications (including reducing salt intake, weight loss, exercise, limiting alcohol, managing stress, and improving sleep)	Further reduce home BP
		Individuals who respond well to this approach are those with hypertension, age ≥65 y, high self-efficacy, and early weight loss (−0.5 kg)

BMI indicates body mass index; BP, blood pressure; and CV, cardiovascular.

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hypertension.[76,77] The first randomized controlled trial of DTx-support for lifestyle modifications (including sodium restriction, weight reduction, increased physical activity, alcohol intake reduction, stress management, and better sleep) in Japanese patients with untreated hypertension (n=390) showed a reduction of 4.3 mm Hg in early morning home SBP in the DTx group compared with standard lifestyle modification alone.[78] The B-INDEX study enrolled 198 patients with hypertension who used DTx to support lifestyle modifications for 6 months and identified as individuals with higher baseline BP, age ≥60 years, greater self-efficacy, successful salt reduction, and modest early weight loss (−0.5 kg within the first 4 weeks) as being more likely to achieve effective BP reduction through DTx.[79]

Morning HBPM is essential for detecting BP changes, including those that occur in response to cold temperatures. In addition, meta-analysis data indicate that the use of HBPM is associated with a reduction in SBP/diastolic BP of −3.3 mm Hg/−1.6 mm Hg, although intensification of antihypertensive drug therapy in people using HBPM could have contributed to this reduction.[80] The reported positive impact of HBPM on BP supports the Japanese Asakatsu BP initiative, encouraging the measurement of morning BP in a variety of everyday settings.[34]

Digital Hypertension

Digital hypertension, in which multidisciplinary teams collaborate using wearable devices, internet-of-things-enabled home BP monitors, cloud-based data platforms, and other digital technologies to comprehensively manage 24-hour BP under real-world conditions, is expected to become a central approach in future hypertension care.[81–85]

We have developed a multisensor home ABPM system capable of simultaneously monitoring environmental information (including indoor temperature, illuminance, humidity, and atmospheric pressure) and integrating these data with physiological parameters (Figure 5).[29,86] This system is characterized by its ability to acquire data on 24-hour variations in central and brachial BP in parallel with environmental factors (eg, temperature) and physical activity. Multiple indoor environmental sensors installed in various rooms of the home (entrance, bedroom, living room, bathroom, toilet), together with an information and communication technology/internet-of-things platform, enable real-time assessment of the indoor microenvironment. A high-sensitivity actigraph provides precise behavioral and positional information, allowing simultaneous acquisition of multimodal data encompassing physiology, behavior, and the environment.

Through the integration of these data, individual-specific BP response profiles, such as actisensitivity (BP reactivity to physical activity), thermosensitivity (BP reactivity to temperature change), and atmosensitivity BP reactivity to atmospheric pressure, can be visualized.[87] Furthermore, multilevel structural modeling allows precise estimation of BP under varying environmental conditions, enabling individualized extraction of seasonal BP profiles and environment-dependent BP responsiveness.

The large-scale time-series data generated by this multisensor BP monitoring system provide a foundation for predicting winter-associated BP elevations and evaluating the effects of environmental interventions. These technologies have the potential to contribute substantially to the practice of anticipation medicine, which aims to anticipate cardiovascular events and guide preemptive intervention.[29,86] The utility of this system for widespread deployment in clinical settings remains to be determined.

PERSPECTIVES AND CONCLUSIONS

Although still a developing field where additional research and evidence are needed, there is a reasonable body of evidence to support BP elevation in response to cold stress. This adds an additional layer of complexity to the management of hypertension during winter months, especially in countries such as Japan, where indoor temperatures during winter are considerably lower than in

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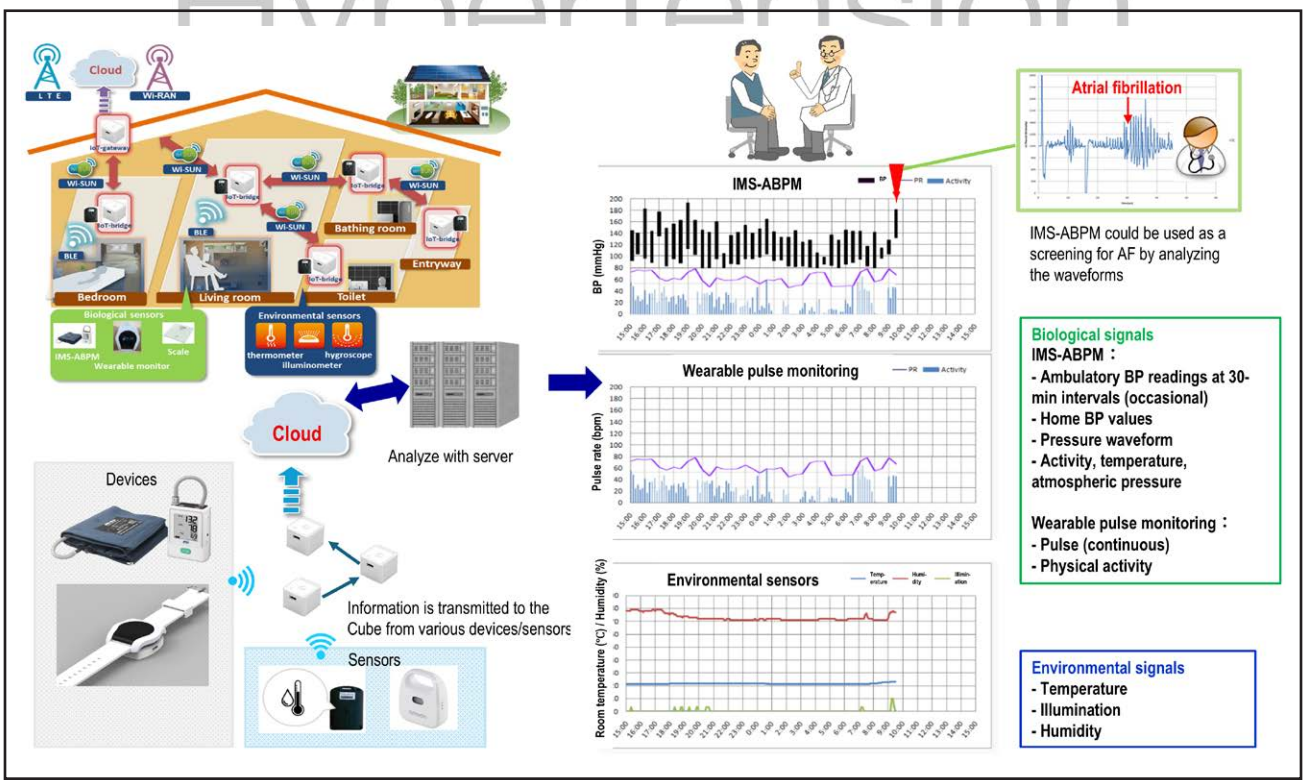

Figure 5. Information and communication technology multisensory (IMS) digital monitoring of blood pressure (BP), physical activity, and the environment.
ABPM indicates ambulatory blood pressure monitoring; AF, atrial fibrillation; IoT, internet of things; PR, pulse rate; and WiSun, WiFi transmission system. Adapted from Kario et al[29] with permission. Copyright ©2017, Elsevier.

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summer and can vary markedly between different parts of the house.

In these settings, it would seem appropriate to take a more comprehensive approach to the management of cold-sensitive hypertension, including optimization of living conditions (especially indoor temperature and temperature along daily movement pathways), clothing, and physical activity. This is particularly relevant for the sensitive period immediately after awakening because of the potential for the BP-elevating effects of low indoor temperature, awakening-related sympathetic activation, and initiation of physical activity to converge. Interventions that encourage awareness of the BP level, such as the more widespread utilization of morning HBPM (as encouraged by the Japanese Asakatsu BP initiative), would seem to be a logical approach to identifying elevated BP and the MBPS, which is the first step to managing these cardiovascular risk factors. Other tools that could be useful in the future include a multisensor home ABPM system, which is currently under development in Japan.

Overall, growing evidence for a contribution of cold-exposure to BP and BP variability suggests the need to take a broader approach to BP control during winter. It is suggested that this should include consideration of the environment, individual behaviors and physiology, along with appropriate management and titration of antihypertensive drug therapy.[88–90] Such an approach has recently been recommended by the Japanese Society of Hypertension: the Morning BP Action in Winter initiative.[5]The aim is to have a comprehensive and actionable strategy that can contribute to reductions in BP and lower cardiovascular morbidity and mortality in the colder months.

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